

Introduction to Quadrotor Unmanned Flight Control – Principles and Applications

Lecture 8: 3D Rotation for Quadrotors

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From 2D to 3D

- In 2D:

θ

One angle describes rotation.

- In 3D:

ϕ, θ, ψ

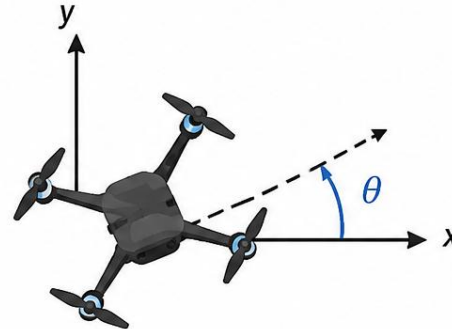
We need three angles:

- Roll
- Pitch
- yaw

2D ROTATION

One angle describes orientation.

θ

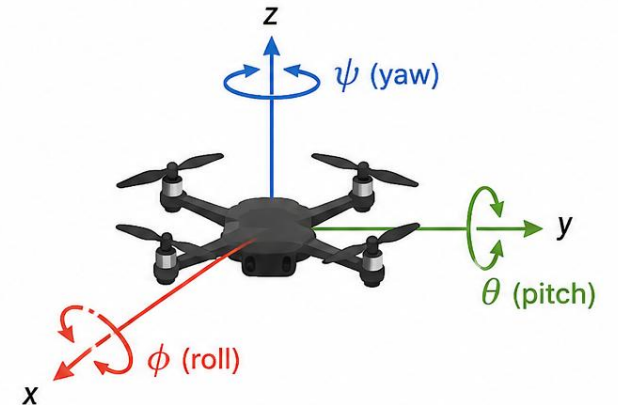


1 variable: θ

3D ROTATION

Three angles describe orientation.

ϕ, θ, ψ



3 variables: ϕ, θ, ψ

3D rotation is harder because orientation is harder to describe.

What Changes from the 2D Model?

2D rotation

$$I\ddot{\theta} = \tau$$

- One angle
- One torque
- One inertia

3D rotation

$$J\ddot{\eta} = \tau + \text{gyroscopic terms}$$

where $\eta = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}$

- Three angles
- Three torques
- Coupling between axes

A Rotation is a Mapping

- A rotation changes how a vector is seen.

$$v_B \rightarrow v_W$$

Body frame vector to world frame vector

- We use a matrix:

$$v_W = R v_B$$

How Do We Describe the Matrix?

- The rotation matrix has 9 numbers.

$$R = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

- But orientation only needs 3 numbers.
- So we parameterize R .
- Most intuitive choice: $\phi, \theta, \psi \rightarrow$ roll, pitch, yaw

Roll, Pitch, Yaw

- **Roll:** ϕ

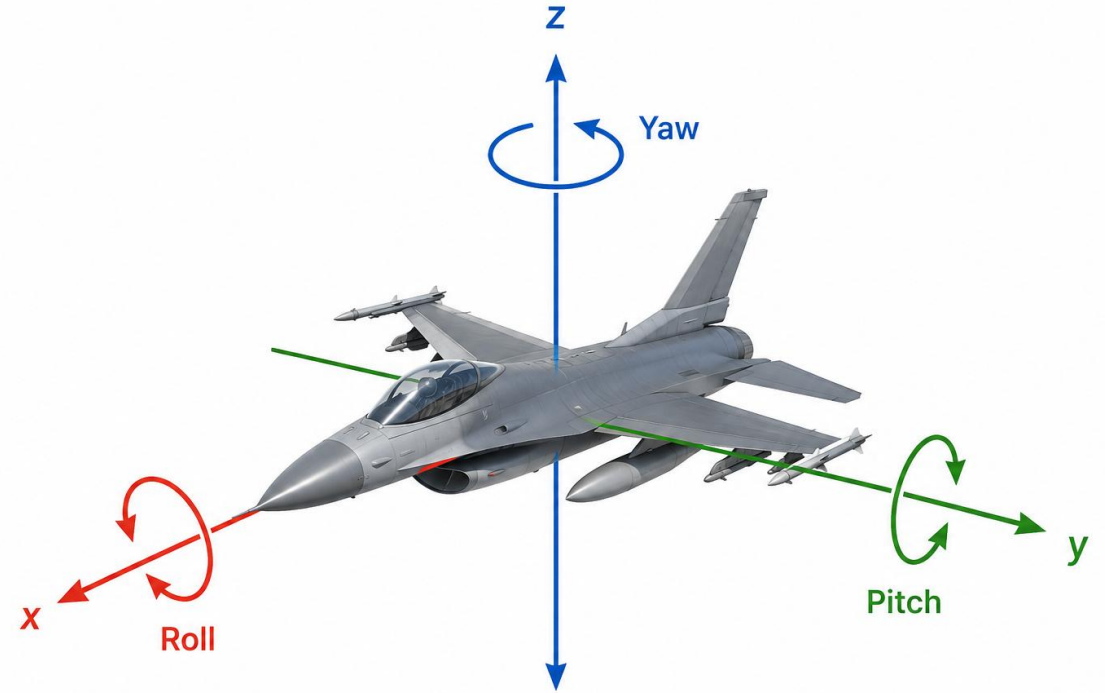
- Rotation about x

- **Pitch:** θ

- Rotation about y

- **Yaw:** ψ

- Rotation about z



Rotation Matrices for Each Axis

- Roll: $R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$

- Pitch: $R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$

- Yaw: $R_z(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Build the Rotation Matrix

Common convention:

$$R = R_z(\psi)R_y(\theta)R_x(\phi)$$

It can be interpreted as:

- Fixed axes (extrinsic): roll $\rightarrow$ pitch $\rightarrow$ yaw
- Body axes (intrinsic): yaw $\rightarrow$ pitch $\rightarrow$ roll

Rotational Dynamics in 3D

- In 2D:

$$I\ddot{\theta} = \tau$$

- In 3D:

$$J\dot{\omega} + \omega \times J\omega = \tau$$

- Torque changes angular velocity.
- Cross-product term appears because 3D rotation is coupled.

Why Euler Angles Are not Perfect

- Euler angles are intuitive. But they have a problem:

$$\theta = \pm 90^\circ$$

- At this point, two rotation axes align.

Gimbal lock

Video

https://www.youtube.com/watch?v=BczeMqU_u2Y